

TECHNICAL PROJECT REPORT

STM32-Based Smart Environmental Monitoring and Control PCB

*Design, Schematic Capture, PCB Layout, and Verification of an STM32F103C8T6-Based
Embedded Controller for Water and Environmental Management*

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Subsystem: Embedded Control PCB — STM32F103C8T6

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Design Tool: eSim

1. Project Overview

1.1 Introduction

The proposed project is a compact embedded control and monitoring system designed around an STM32F103C8T6 microcontroller. The system integrates multiple sensing, communication, storage, display, and actuator interfaces on a single printed circuit board (PCB). The circuit is intended for an automated environmental and water-management application in which sensor information is acquired by the microcontroller and used to monitor conditions and control external devices such as water and air pumps.

The schematic developed for this project is organized into the following functional modules:

- 3.3 V regulated power supply
- STM32F103C8T6 microcontroller and 16 MHz clock circuit
- Water pump MOSFET driver
- Air pump MOSFET driver
- OLED display interface (I²C)
- Micro-SD card storage interface
- pH sensor interface
- Turbidity sensor interface
- Temperature sensor interface
- Ultrasonic water-level sensor interface
- Micro-USB interface
- UART debugging interface
- I²C expansion connector
- SWD programming and debugging interface

This modular architecture allows the PCB to function as a single centralized controller capable of supporting several external sensors and peripherals simultaneously, reducing the wiring, size, and integration overhead typically associated with multi-board prototype systems.

1.2 Motivation

Environmental and water-management monitoring systems commonly require several sensors and actuators operating simultaneously — for example, water quality sensors, level sensors, and pump actuators. Implementing each of these functions on separate development boards increases wiring complexity, physical footprint, power consumption, and the number of potential points of failure.

The motivation for this design is to consolidate all required sensing, control, storage, and communication functions onto a single dedicated PCB. The STM32F103C8T6 was selected because it provides sufficient

GPIO count, ADC channels, communication peripherals (UART, I²C, USB), timers for PWM generation, and processing capability to integrate all of the required peripherals on one board.

1.3 Objectives

- Design a compact PCB centered on the STM32F103C8T6 microcontroller.
- Provide a regulated 3.3 V supply for all digital and analog electronics.
- Interface environmental and water-quality sensors (pH, turbidity, temperature, ultrasonic level) with the microcontroller.
- Provide dedicated low-side MOSFET driver circuits for the water and air pumps.
- Support local, non-volatile data storage through a Micro-SD card.
- Provide a display interface (OLED/I²C) and a general-purpose I²C expansion port.
- Provide USB, UART, and SWD connectivity for programming, debugging, and diagnostics.
- Develop a reliable PCB layout suitable for practical embedded-system operation.
- Physically and electrically separate sensitive low-voltage analog signal circuitry from higher-current actuator switching circuitry.

2. Circuit Description

2.1 Overall Circuit Architecture

The STM32F103C8T6 microcontroller forms the central processing unit of the system. Sensor signals — pH, turbidity, temperature, and ultrasonic water-level — are supplied to the MCU through dedicated analog and digital input interfaces, while PWM control outputs operate the water-pump and air-pump MOSFET driver circuits. Communication interfaces (UART, USB, I²C) and the SWD programming port provide access to external development and diagnostic tools. Regulated 3.3 V power is distributed from a single linear-regulator supply to the MCU core, VDDA analog rail, and all peripheral interfaces.

Conceptually, the system can be represented as a central controller (STM32) surrounded by four functional groups: sensing (pH, turbidity, temperature, ultrasonic), actuation (water pump, air pump), communication/programming (UART, USB, I²C, SWD), and storage/display (Micro-SD, OLED).

2.2 3.3 V Power Supply

The power supply section converts the incoming VBUS/+5 V input to a regulated +3.3 V rail using an AMS1117-3.3 linear voltage regulator (U1). Bulk input and output capacitors (C1, C2 — 22 μ F each) improve supply stability and reduce voltage fluctuations under transient load. A red LED (D1) with a series current-limiting resistor (R1, 1.5 k Ω) provides visual confirmation that the regulated 3.3 V supply is present. The resulting rail supplies the STM32 and the majority of peripheral interfaces on the board.

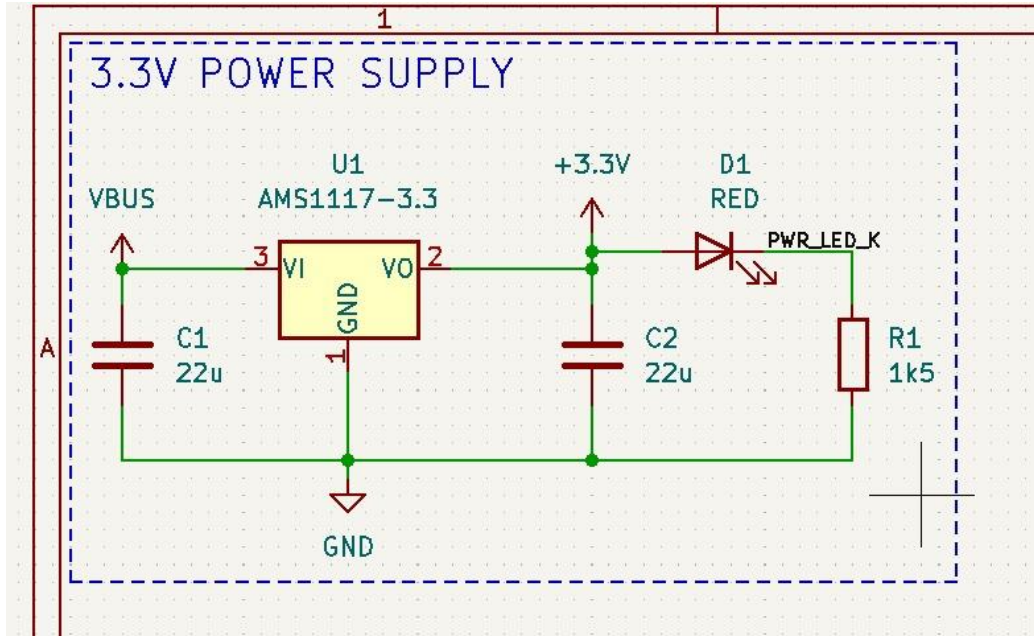


Figure 2.1 — 3.3 V linear power supply section (U1: AMS1117-3.3, C1/C2 bulk capacitors, D1/R1 power-indicator LED).

2.3 STM32F103C8T6 Microcontroller and Clock Circuit

The main controller is the STM32F103C8T6, an ARM Cortex-M3-based microcontroller in an LQFP-48 package. It serves as the central processing and control unit, acquiring sensor data and generating the corresponding actuator control signals. As shown in the schematic, GPIO pins are allocated to pH ADC input, turbidity ADC input, temperature data, ultrasonic trigger/echo, SD-card SPI signals, OLED I²C signals, water- and air-pump PWM outputs, USB D+/D–, and SWDIO/SWCLK.

A 16 MHz crystal oscillator (Y1) is connected to the MCU high-speed external oscillator pins (PD0/PD1, HSE_IN/HSE_OUT), with two 10 pF load capacitors (C12, C13) forming the standard crystal oscillator network. The external crystal provides a stable timing reference for the CPU core and for peripheral communication (UART, I²C, USB, SPI).

Reset and boot configuration are provided through the NRST pin (with 100 nF capacitor C11) and the BOOT0 pin, which is set through a single-pole double-throw slide switch (SW1) and a 10 kΩ resistor (R2), allowing selection between normal boot-from-flash operation and system-memory bootloader mode.

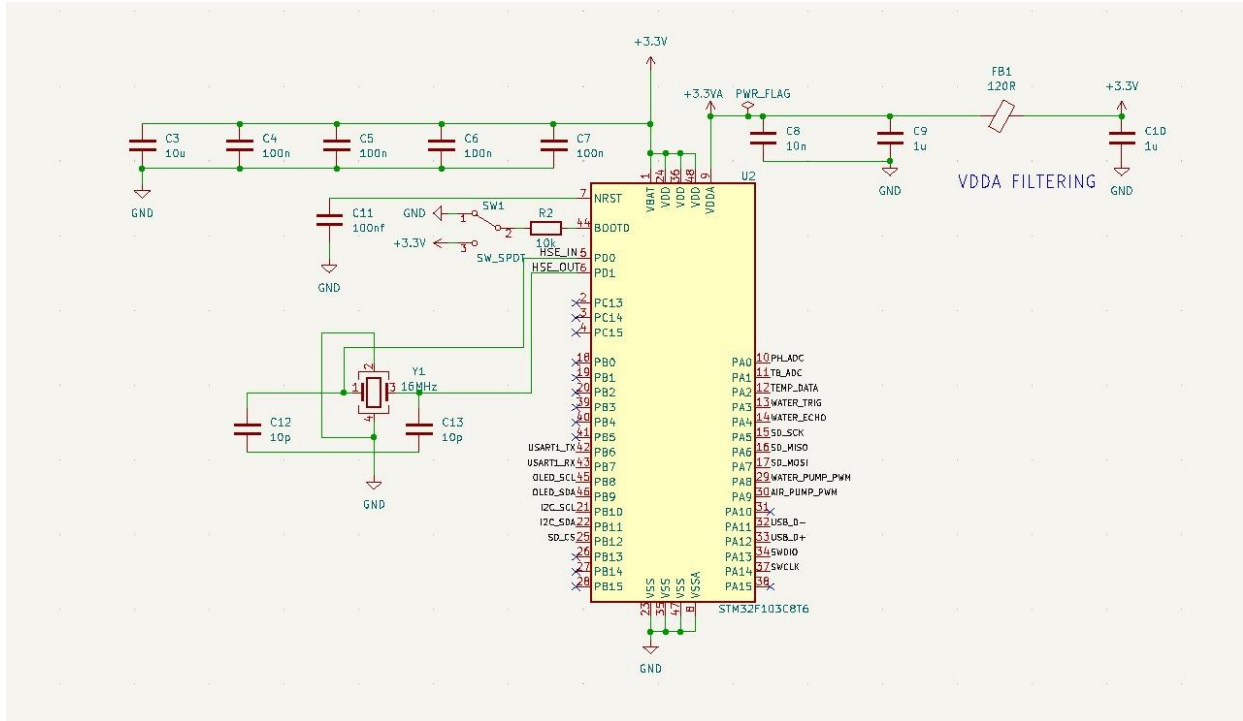


Figure 2.2 — STM32F103C8T6 controller, decoupling network, NRST/BOOT0 configuration, 16 MHz crystal oscillator, and VDDA analog-supply filtering (FB1 ferrite bead).

2.4 VDDA Analog Supply Filtering

The analog supply of the microcontroller (VDDA) is filtered separately from the main digital 3.3 V rail. A 120 Ω ferrite bead (FB1) separates the analog supply path from the digital +3.3 V rail, with 10 nF and 1 μF capacitors (C8, C9) placed on the filtered side. This arrangement reduces high-frequency digital switching noise from reaching the MCU's internal ADC reference and analog comparator circuitry — an important consideration because the board includes analog sensor inputs such as the pH and turbidity sensors.

2.5 Water Pump Driver

The water-pump section controls an external pump supplied from an independent +12 V rail (connector J9, PUMP_PWR). The STM32 generates a WATER_PUMP_PWM control signal that drives the gate of an AO3400A N-channel MOSFET (Q1) through a 100 Ω series gate resistor (R13). A 10 kΩ pull-down resistor (R14) ensures the MOSFET gate is held low — and the MOSFET remains OFF — whenever the control signal is floating, such as during MCU reset or power-up. The MOSFET performs low-side switching, connecting the pump return path to ground when energized. A 1N5822 Schottky diode (D2) is connected across the pump output to clamp the inductive flyback voltage generated when the pump current is switched off, protecting Q1 from transient overvoltage.

2.6 Air Pump Driver

The air-pump driver mirrors the water-pump circuit topology. The STM32 provides an AIR_PUMP_PWM signal to a second AO3400A MOSFET (Q2) through gate resistor R15 (100 Ω) and pull-down resistor R16

(10 k Ω), with a 1N5822 flyback diode (D3) protecting the switch. Implementing separate, independent MOSFET drivers for the two pumps allows the water pump and air pump to be switched or PWM-modulated independently, since the two actuators may be required to operate on different schedules or at different duty cycles.

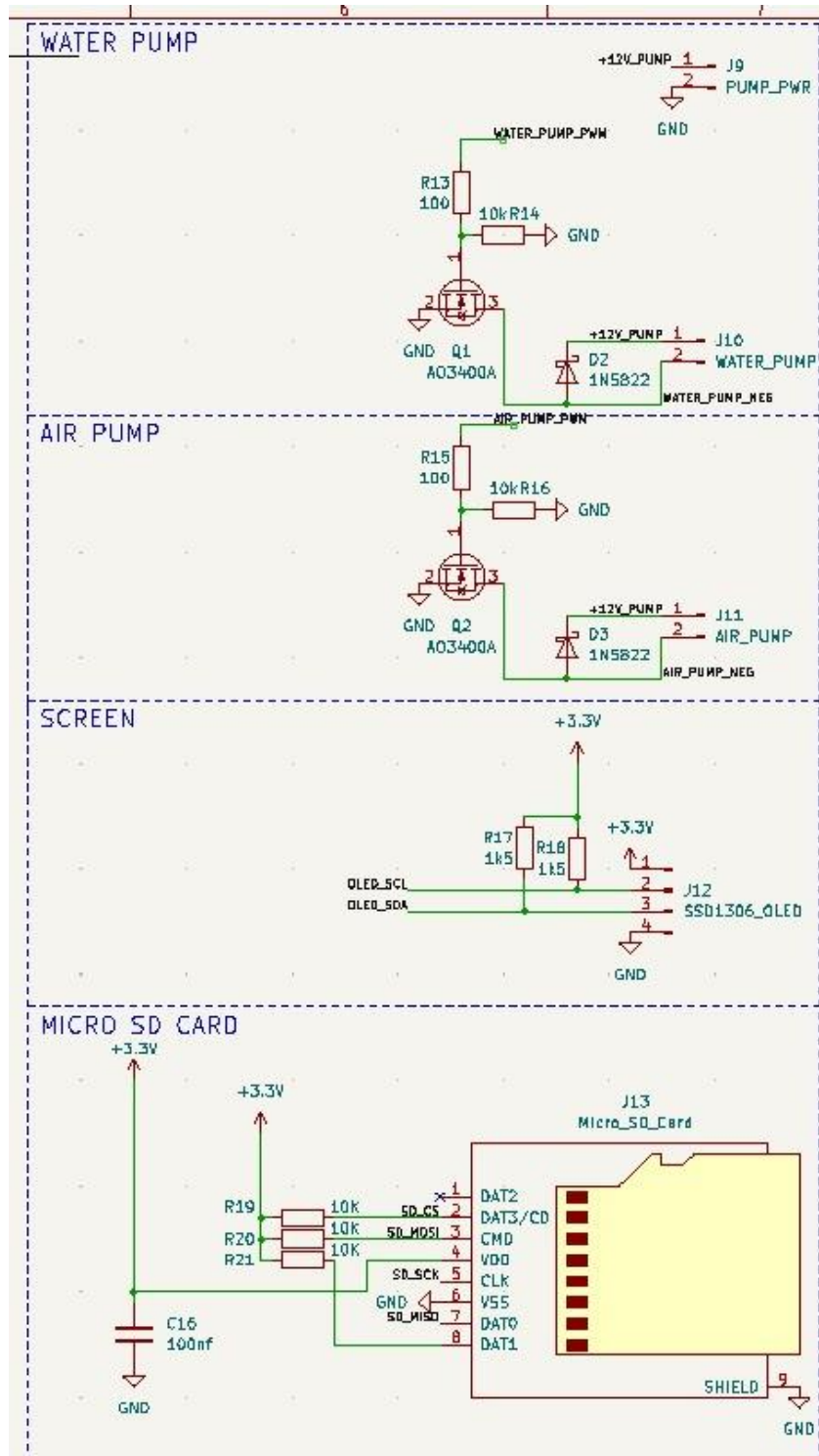


Figure 2.3 — Water-pump driver (Q1/R13/R14/D2), air-pump driver (Q2/R15/R16/D3), OLED display connector, and Micro-SD card interface.

2.7 Display Interface

A four-pin connector (J12) provides +3.3 V, GND, and I²C SDA/SCL signals for an SSD1306-type OLED display module. Pull-up resistors R17 and R18 (1.5 k Ω) are placed on the OLED_SCL and OLED_SDA lines respectively to satisfy the open-drain I²C bus requirements. The display can present real-time measurements such as pH, temperature, turbidity, and water level, as well as pump operating status.

2.8 I²C Expansion Interface

An independent four-pin I²C expansion connector (J2) exposes +3.3 V, GND, I2C_SCL, and I2C_SDA, with dedicated pull-up resistors R3 and R4 (1.5 k Ω). This connector allows additional I²C-compatible sensors or peripheral modules to be added to the system without requiring further dedicated MCU pins, improving the expandability of the design.

2.9 Micro-SD Card Interface

A Micro-SD card socket (J13) provides non-volatile local data storage over an SPI-mode interface (SD_CS, SD_MOSI, SD_SCK, SD_MISO), with series/pull resistors R19, R20, R21 (10 k Ω) used for signal conditioning and level compatibility, and a 100 nF decoupling capacitor (C16) at the card supply pin. The SD card can be used to log sensor measurements, pump operating states, and time-stamped environmental data for later analysis.

2.10 UART, I²C Expansion, SWD, USB, and pH Sensor Interfaces

A dedicated four-pin UART connector (J1) exposes USART1_TX and USART1_RX for serial debugging, allowing the firmware to stream sensor readings, error messages, and calibration data to a host PC during development. A four-pin SWD connector (J3) exposes SWDIO and SWCLK together with +3.3 V and GND, enabling direct firmware programming and low-level debugging without requiring a separate programming header to be added after assembly.

A Micro-USB connector (J4) provides VBUS, D+, D–, and GND, with a 1.5 k Ω resistor (R5) on the VBUS sense path; this connector can be used for board power and/or USB communication depending on the firmware implementation. The pH sensor connector (J5) is filtered by a 100 nF capacitor (C14) and conditioned by resistor R6, with its analog output (PH_ADC) routed to an ADC-capable STM32 input; the firmware converts the measured voltage into a pH value using a calibration curve.

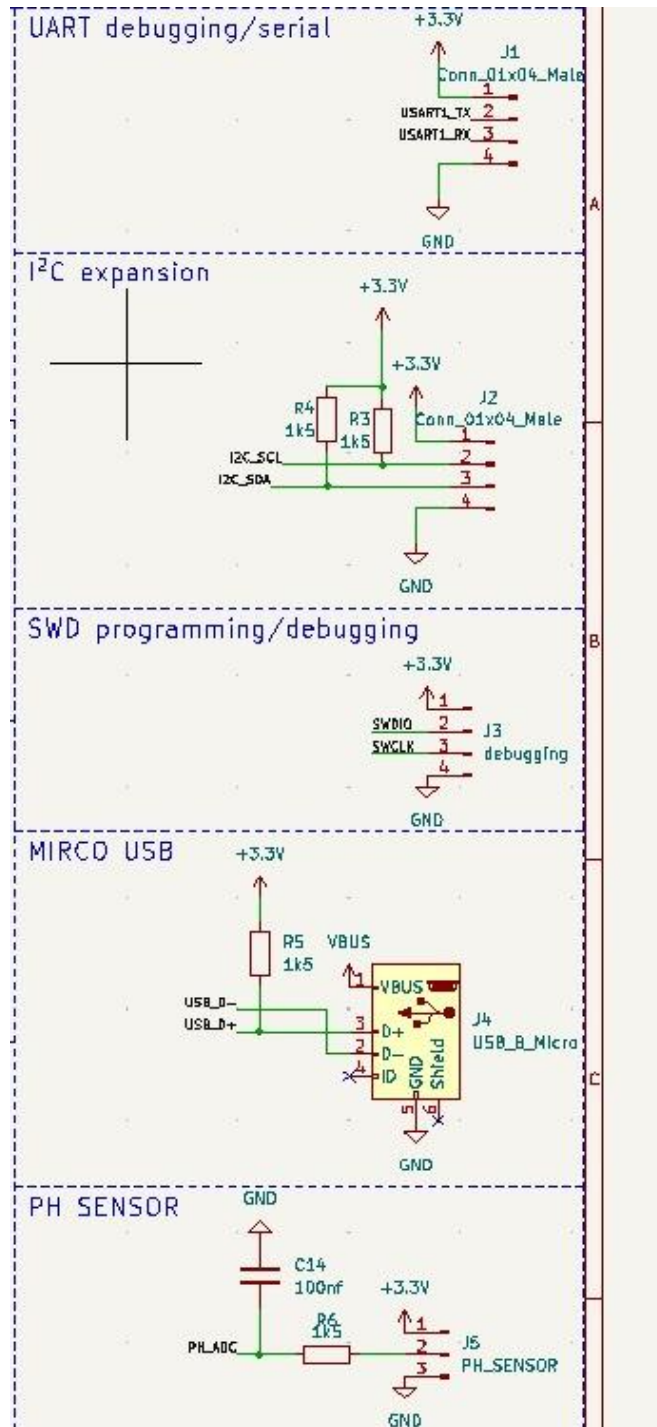


Figure 2.4 — UART debugging connector (J1), I²C expansion port (J2, R3/R4 pull-ups), SWD programming connector (J3), Micro-USB interface (J4), and pH sensor interface (J5, R6/C14).

2.11 Turbidity, Temperature, and Ultrasonic Water-Level Interfaces

The turbidity-module interface (J6) uses a resistive divider (R7 = 12 k Ω , R8 = 27 k Ω) together with a 1.5 k Ω resistor (R9) and a 100 nF filter capacitor (C15) to condition the sensor's analog output (TB_ADC) before it reaches the MCU ADC input.

The temperature-sensor connector (J7) provides +3.3 V, GND, and a data line (TEMP_DATA) pulled through a 4.8 k Ω resistor (R10), suitable for a single-wire digital temperature sensor.

The ultrasonic water-level sensor connector (J8) is supplied from +5 V and provides trigger (WATER_TRIG) and echo (WATER_ECHO) signal lines, conditioned through resistors R11 (15 k Ω) and R12 (10 k Ω) which form a divider to bring the 5 V echo signal down to a level compatible with the 3.3 V STM32 input. The ultrasonic sensor determines the distance to the water surface; the firmware converts this distance into a water-level value using known tank geometry, giving a non-contact level-measurement method.

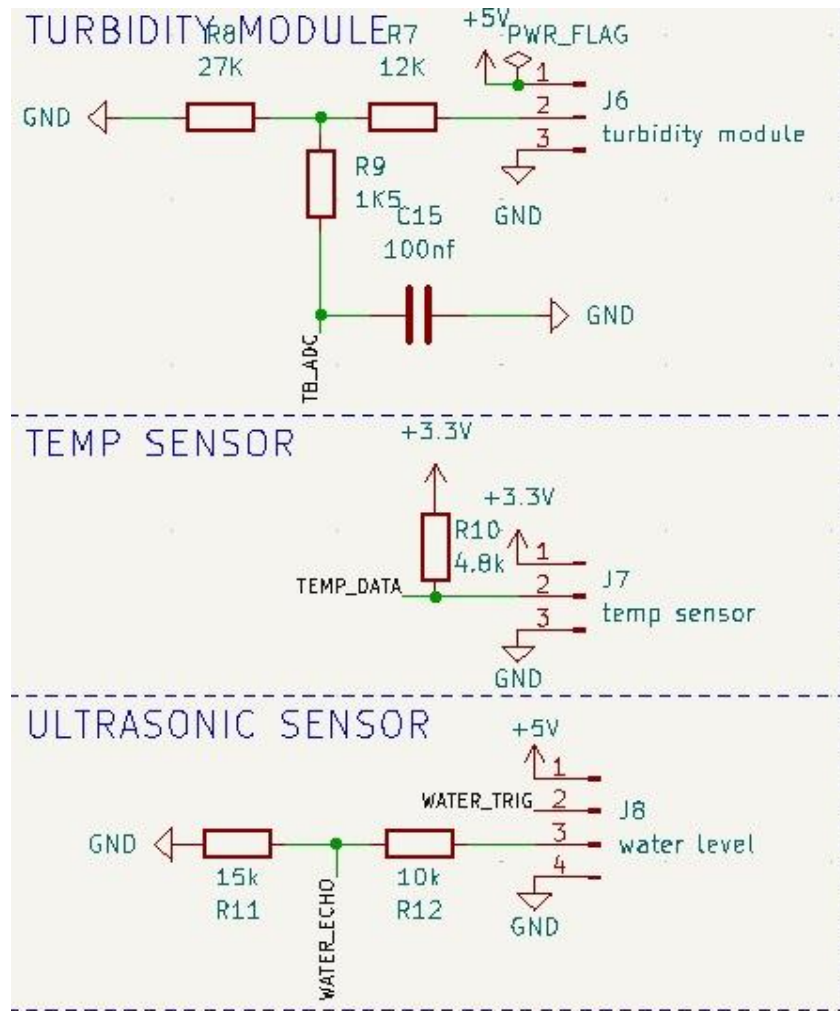


Figure 2.5 — Turbidity-module analog front end (R7–R9, C15), temperature-sensor connector (R10), and ultrasonic water-level sensor interface (R11, R12).

3. Schematic Screenshots

The complete schematic was captured in eSim and divided into functional sheets/blocks, consistent with the circuit description in Section 2. The figures below summarize, for reference, where each functional block appears in the schematic set.

Figure	Schematic Block	Key Reference Designators
Fig. 2.1	3.3 V power supply	U1, C1, C2, D1, R1
Fig. 2.2	MCU core, clock, reset/boot, VDDA filtering	U2, Y1, C3–C13, FB1, SW1, R2
Fig. 2.3	Water pump, air pump, OLED, Micro-SD	Q1, Q2, D2, D3, R13–R16, J9–J13
Fig. 2.4	UART, I ² C expansion, SWD, USB, pH sensor	J1–J5, R3–R6
Fig. 2.5	Turbidity, temperature, ultrasonic sensors	J6–J8, R7–R12

Refer to Figures 2.1 through 2.5 in Section 2 for the annotated schematic captures; they are not repeated here to avoid duplication.

4. Footprint Assignment Details

Symbol : Footprint Assignments		
1	C1 -	22u : Capacitor_SMD:C_0805_2012Metric
2	C2 -	22u : Capacitor_SMD:C_0805_2012Metric
3	C3 -	10u : Capacitor_SMD:C_0603_1608Metric
4	C4 -	100n : Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder
5	C5 -	100n : Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder
6	C6 -	100n : Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder
7	C7 -	100n : Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder
8	C8 -	10n : Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder
9	C9 -	1u : Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder
10	C10 -	1u : Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder
11	C11 -	100nf : Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder
12	C12 -	10p : Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder
13	C13 -	10p : Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder
14	C14 -	100nf : Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder
15	C15 -	100nf : Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder
16	C16 -	100nf : Capacitor_SMD:C_0402_1005Metric
17	D1 -	RED : LED_SMD:LED_0603_1608Metric
18	D2 -	1N5822 : Diode_THT:D_DO-201AD_P15.24mm_Horizontal
19	D3 -	1N5822 : Diode_THT:D_DO-201AD_P15.24mm_Horizontal
20	FB1 -	120R : Inductor_SMD:L_0603_1608Metric
21	J1 -	Conn_01x04_Male : Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Horizontal
22	J2 -	Conn_01x04_Male : Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Horizontal
23	J3 -	debugging : Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Horizontal
24	J4 -	USB_B_Micro : Connector_USB:USB_Micro-B_Wuerth_629105150521
25	J5 -	PH_SENSOR : Connector_PinHeader_2.54mm:PinHeader_1x03_P2.54mm_Vertical
26	J6 -	turbidity module : Connector_PinHeader_2.54mm:PinHeader_1x03_P2.54mm_Vertical
27	J7 -	temp sensor : Connector_PinHeader_2.54mm:PinHeader_1x03_P2.54mm_Vertical
28	J8 -	water level : Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Horizontal
29	J9 -	PUMP_FWR : Connector_Phoenix_MSTB:PhoenixContact_MSTBVA_2,5_2-G_1x02_P5.00mm_Vertical
30	J10 -	WATER_PUMP : Connector_Phoenix_MSTB:PhoenixContact_MSTBVA_2,5_2-G_1x02_P5.00mm_Vertical
31	J11 -	AIR_PUMP : Connector_Phoenix_MSTB:PhoenixContact_MSTBVA_2,5_2-G_1x02_P5.00mm_Vertical
32	J12 -	SSD1306_OLED : Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical
33	J13 -	Micro_SD_Card : Connector_Card:microSD_HC_Molex_47219-2001
34	Q1 -	AO3400A : Package_TO_SOT_SMD:SOT-23
35	Q2 -	AO3400A : Package_TO_SOT_SMD:SOT-23
36	R1 -	1k5 : Resistor_SMD:R_0603_1608Metric

Figure 4.1 — Symbol–footprint assignment table (part 1 of 2): power, decoupling, MOSFETs, connectors, and the Micro-SD/USB/pin-header components.

36	R1	-	1k5	: Resistor_SMD:R_0603_1608Metric
37	R2	-	10k	: Resistor_SMD:R_0603_1608Metric
38	R3	-	1k5	: Resistor_SMD:R_0603_1608Metric
39	R4	-	1k5	: Resistor_SMD:R_0603_1608Metric
40	R5	-	1k5	: Resistor_SMD:R_0603_1608Metric
41	R6	-	1k5	: Resistor_SMD:R_0603_1608Metric
42	R7	-	12K	: Resistor_SMD:R_0603_1608Metric
43	R8	-	27K	: Resistor_SMD:R_0603_1608Metric
44	R9	-	1K5	: Resistor_SMD:R_0603_1608Metric
45	R10	-	4.8k	: Resistor_SMD:R_0603_1608Metric
46	R11	-	15k	: Resistor_SMD:R_0603_1608Metric
47	R12	-	10k	: Resistor_SMD:R_0603_1608Metric
48	R13	-	100	: Resistor_SMD:R_0603_1608Metric
49	R14	-	10k	: Resistor_SMD:R_0603_1608Metric
50	R15	-	100	: Resistor_SMD:R_0603_1608Metric
51	R16	-	10k	: Resistor_SMD:R_0603_1608Metric
52	R17	-	1k5	: Resistor_SMD:R_0603_1608Metric
53	R18	-	1k5	: Resistor_SMD:R_0603_1608Metric
54	R19	-	10K	: Resistor_SMD:R_0603_1608Metric
55	R20	-	10K	: Resistor_SMD:R_0603_1608Metric
56	R21	-	10K	: Resistor_SMD:R_0603_1608Metric
57	SW1	-	SW_SPDT	: Button_Switch_THT:SW_E-Switch_EG1224_SPDT_Angled
58	U1	-	AMS1117-3.3	: Package_TO_SOT_SMD:SOT-223-3_TabPin2
59	U2	-	STM32F103C8T6	: Package_OFF:LQFP-48_7x7mm_P0.5mm
60	Y1	-	16MHz	: Crystal:Crystal_SMD_3225-4Pin_3.2x2.5mm

Figure 4.2 — Symbol–footprint assignment table (part 2 of 2): resistors, the SPDT switch, the AMS1117 regulator, the STM32F103C8T6, and the crystal.

5. PCB Layout Screenshots

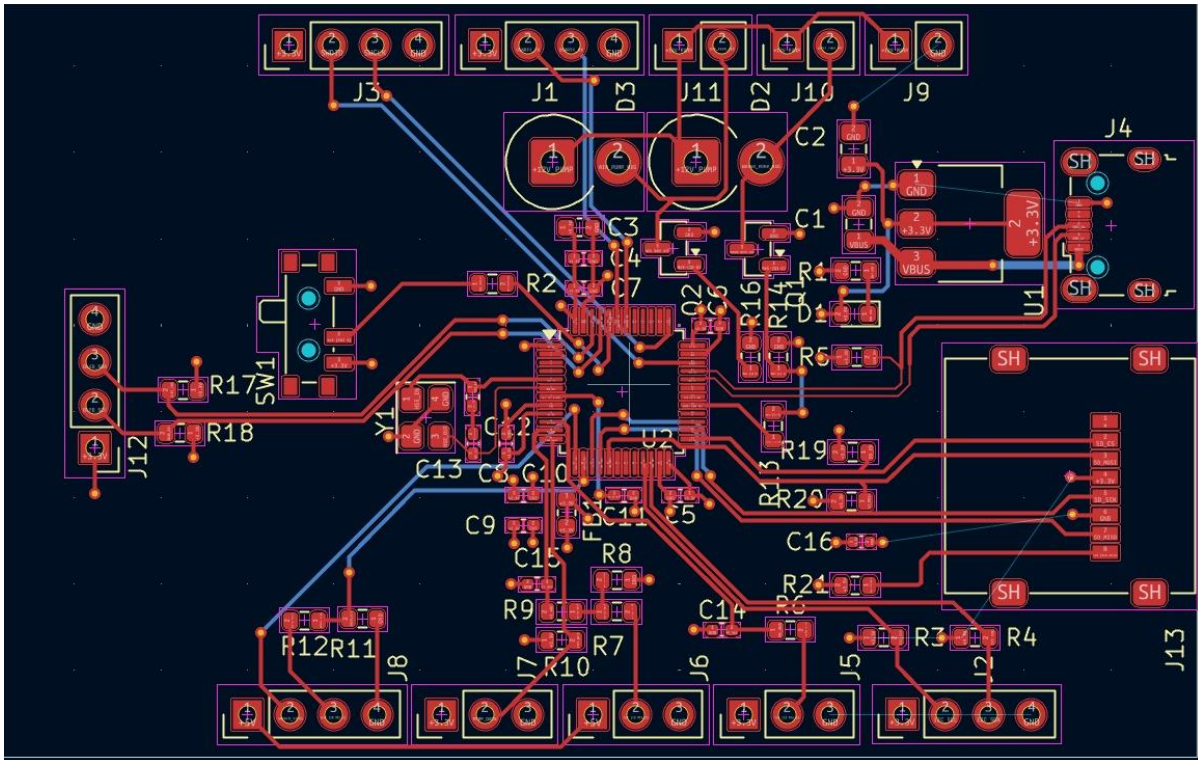


Figure 5.1 — Top-copper PCB routing. The MCU (U2) is placed centrally; decoupling capacitors surround its power pins; the crystal (Y1) sits close to the oscillator pins; the pump MOSFETs (Q1/Q2) and flyback diodes are grouped near their respective connectors (J9–J11); connectors for sensors, UART, I²C, SWD, and USB are placed along the board periphery.

Critical placement and routing decisions visible in the layout include:

- Decoupling capacitors (C3–C7) placed immediately adjacent to the MCU VDD pins.
- Crystal Y1 and its load capacitors (C12, C13) kept close to the MCU HSE pins with short, direct traces.
- VDDA filtering components (FB1, C8, C9) placed close to the MCU analog-supply pin.
- Pump MOSFETs Q1/Q2 and diodes D2/D3 placed close to connectors J9–J11 to minimize the high-current loop.
- USB-related components (R5) kept close to the Micro-USB connector J4.
- Micro-SD card traces (SD_CS/MOSI/MISO/SCK) kept short between U2 and J13.
- Sensor connectors (J5–J8) positioned along the board edge for accessible external wiring.

6. Routing Methodology

6.1 Layout Strategy and Tools

The PCB was designed and routed in eSim, using a four-layer copper stack-up: Layer 1 (top) and Layer 4 (bottom) carry components and signal routing, while Layer 2 and Layer 3 are dedicated internal GND and +3.3 V planes respectively. Routing was performed primarily by hand to allow direct control over trace placement relative to noise-sensitive analog signals and high-current actuator paths; interactive router assistance within eSim was used to speed up routine digital-signal connections, followed by manual clean-up of the identified critical nets.

6.2 Component Placement

Placement was carried out according to signal-flow and electrical-noise considerations, following the strategy outlined in Section 5: the MCU centrally located, decoupling/clock/VDDA-filter components placed immediately adjacent to their associated MCU pins, pump drivers grouped near their output connectors, and sensor/communication connectors distributed along the board edges.

6.3 Power Routing

The +3.3 V supply net was routed using wide copper traces sized for the expected peripheral current, and ground was implemented as a continuous copper region wherever possible to minimize return-path impedance. Because the pump supply (+12 V, from J9) is electrically distinct from the MCU's low-current logic supply, the high-current pump paths were routed separately from the low-current sensor and logic paths, following the general rule that high-current actuator paths should be short and wide, while sensitive analog-signal paths should be short and isolated from noisy nets.

6.4 Digital and Analog Signal Routing

I²C signal pairs (SDA/SCL) for both the OLED interface and the expansion connector were routed with reasonably short trace lengths and kept away from the pump-switching nodes. The USB D+/D– pair was routed as a short, direct pair with minimal vias, consistent with controlled-impedance practice for

differential USB signaling. The crystal-oscillator traces (Y1 to PD0/PD1) were kept extremely short and were routed away from the pump-driver switching nodes to avoid coupling noise into the MCU clock reference.

The pH-sensor signal path (PH_ADC), being the most noise-sensitive net on the board, was given a short, direct trace from connector J5 to the MCU ADC input, routed away from the MOSFET drain nodes and the +12 V pump-supply traces, with a clean, unbroken ground return underneath and a minimum number of vias, consistent with the analog-routing guidance in Section 2.4.

6.5 Grounding Strategy

A common ground reference is used for the low-voltage electronics, implemented as a continuous ground copper region to minimize return-path impedance and reduce electromagnetic interference. Return currents associated with the water and air pumps were kept from flowing through the narrower portions of the ground network serving the pH and turbidity sensor circuitry, to prevent switching-current-induced ground bounce from corrupting the low-level analog measurements.

7. DRC Results

Prior to layout finalization, the design was verified using KiCad's Electrical Rules Checker (ERC) at the schematic level and the Design Rules Checker (DRC) at the PCB level. The ERC pass — checking for unconnected pins, conflicting pin types, and missing power-flag symbols — is shown below.

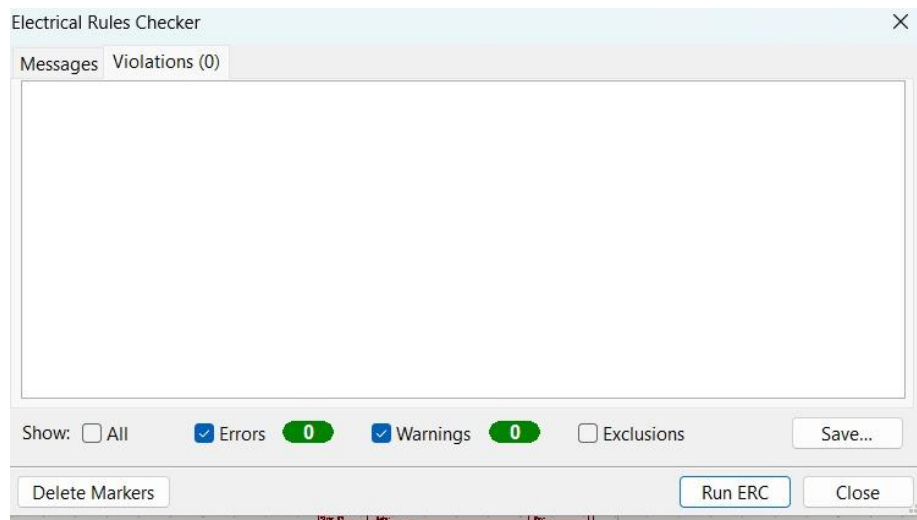


Figure 7.1 — Electrical Rules Checker (ERC) results: 0 errors and 0 warnings reported across the schematic.

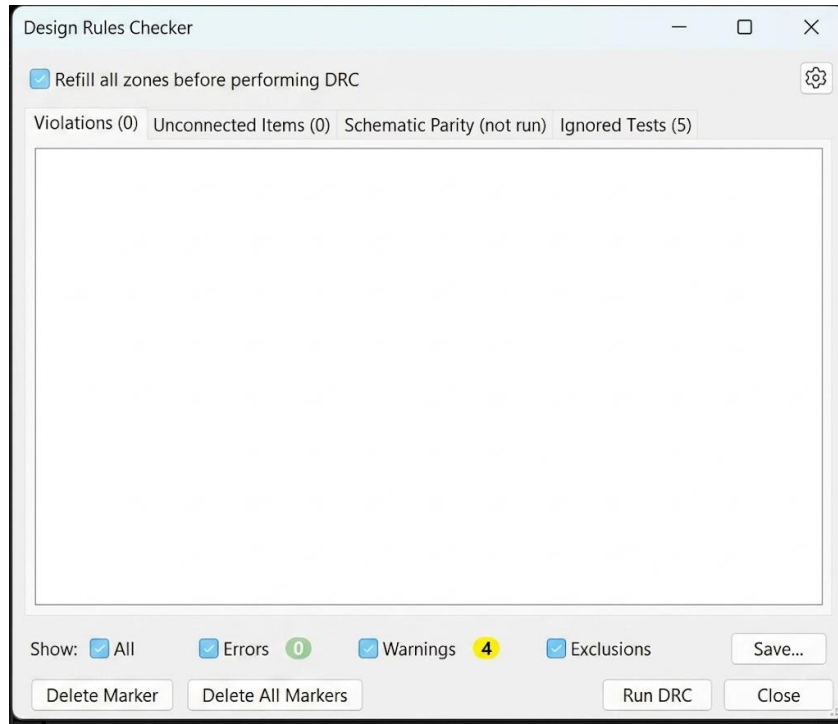


Figure 7.2 — PCB-level Design Rules Checker (DRC) results, run with "Refill all zones before performing DRC" enabled: the Violations tab lists 0 items and the Unconnected Items tab lists 0 items. Four warning-class checks were excluded from the active rule set (visible under the Warnings filter count) and five tests are marked under Ignored Tests; Schematic Parity was not run for this pass.

No unresolved DRC violations or unconnected items remained in the final routed layout. Any clearance or track-width flags raised during intermediate routing passes were corrected by adjusting trace width or spacing in the affected area before the design was considered complete. As noted in Section 9.6, a full physical DRC report and post-fabrication inspection remain recommended steps prior to production release.

8. 3D View Screenshots

KiCad's 3D viewer was used to render the completed board for visual verification of component placement, footprint orientation, and mechanical fit (e.g., OLED module standoffs, Micro-USB and Micro-SD connector clearance).

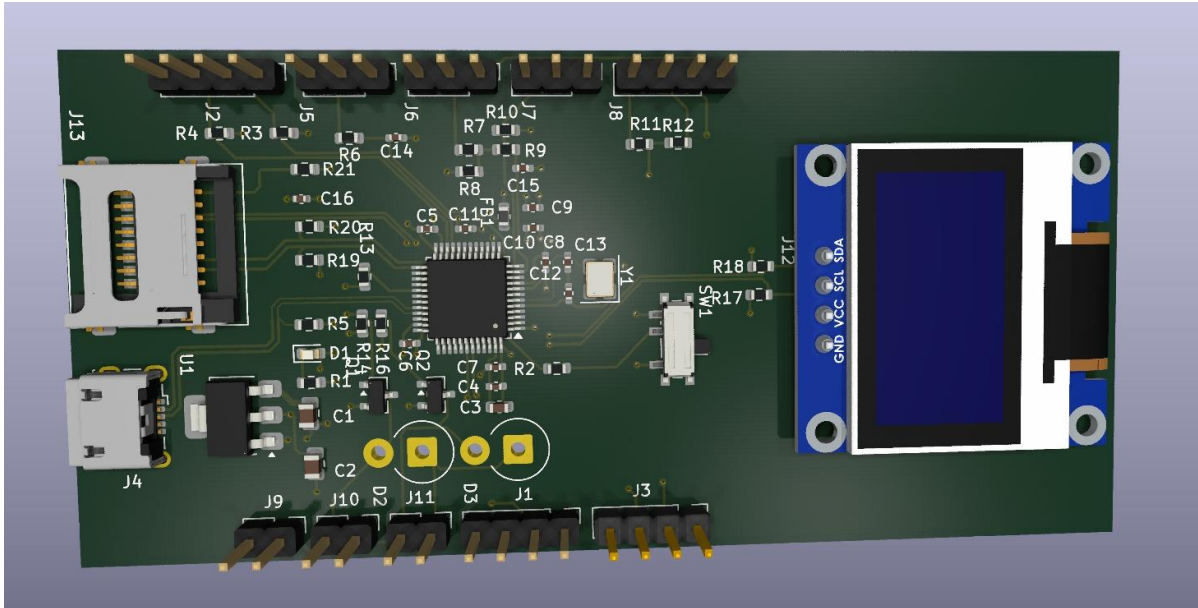


Figure 8.1 — 3D top-side render. Visible: STM32F103C8T6 (U2), crystal (Y1), Micro-SD socket (J13), Micro-USB connector (J4), AMS1117 regulator (U1), pump MOSFETs (Q1/Q2) and flyback diodes (D2/D3), boot-select switch (SW1), OLED display module (J12), and the UART/I²C/SWD/sensor pin headers along the board edges.

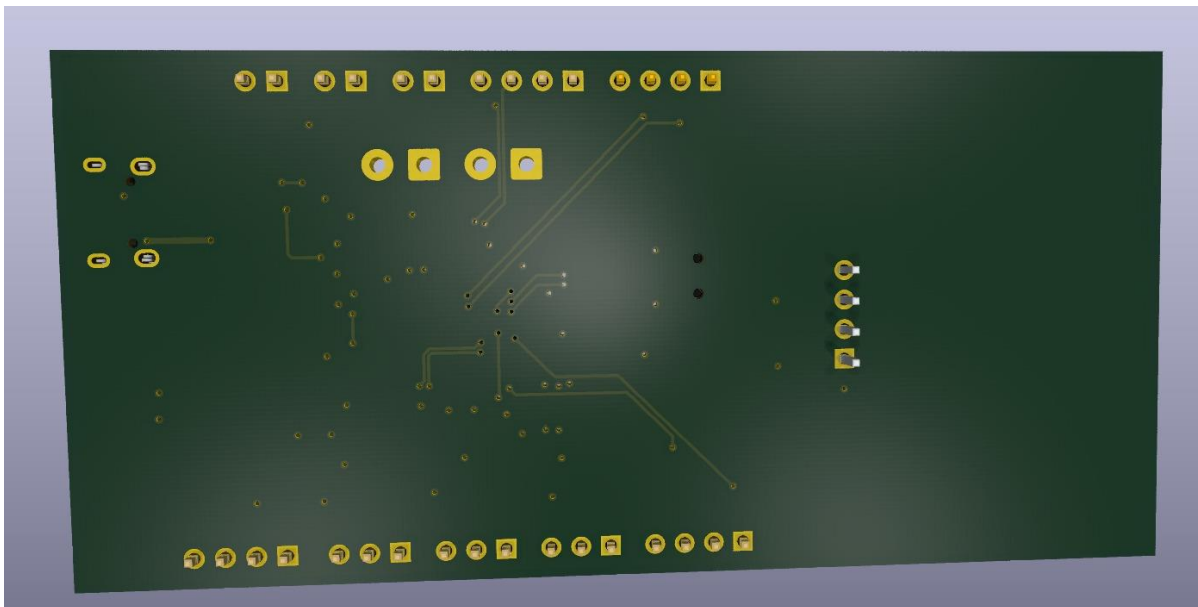


Figure 8.2 — 3D bottom-side render, showing the bottom-copper routing and via placement beneath the connector footprints.

9. Design Considerations and Conclusions

9.1 Performance

The STM32F103C8T6 provides a suitable central platform for coordinating multiple sensors and actuators simultaneously. The architecture supports concurrent sensor acquisition, data processing, display output, data logging, pump control, and USB/UART/SWD communication, making the board considerably more capable than a single-purpose controller.

9.2 Power Consumption

The design separates low-power digital electronics (STM32 and sensor interfaces, powered from the regulated 3.3 V rail) from the higher-current pump actuators (powered from an independent +12 V supply). This separation reduces the risk of pump-current transients affecting the microcontroller supply. The AMS1117 linear regulator is simple and low-cost but less efficient than a switching regulator when the input-to-output voltage differential is large; thermal performance should therefore be considered when selecting the actual input supply voltage and expected peripheral current.

9.3 Cost Considerations

The design uses widely available, low-cost components (AMS1117 regulator, AO3400A MOSFETs, standard passive components, the STM32F103C8T6, and standard connectors), keeping the bill of materials economical. Consolidating all functions onto a single PCB, rather than multiple separate development boards, also reduces PCB area, connector count, wiring, and overall system cost.

9.4 Safety

The pump drivers introduce higher-current, inductive loads. Flyback protection diodes (D2, D3) across the pump outputs reduce voltage spikes generated when pump current is switched off. Adequate clearance and trace width were used for the external 12 V power connections, and the USB and low-voltage signal circuitry were kept electrically separated from the noisier actuator paths as far as practical.

9.5 Reliability

- Decoupling: Multiple capacitors around the MCU supply pins suppress high-frequency supply disturbances.
- Analog filtering: The VDDA supply is isolated using a ferrite bead and dedicated filter capacitors.
- Gate control: Pull-down resistors on both pump-MOSFET gates ensure a defined OFF state during MCU reset or power-down.
- Debugging interfaces: UART and SWD connectors simplify fault diagnosis and firmware development.
- Modular connectors: Separate connectors for each sensor and actuator simplify maintenance and field replacement.

9.6 Conclusion

The developed PCB architecture provides a comprehensive embedded control platform based on the STM32F103C8T6 microcontroller. The system combines environmental sensing (pH, turbidity, temperature), water-level monitoring, data storage, display functionality, communication interfaces, and independent actuator control within a unified two-layer PCB design. Dedicated MOSFET drivers allow the water and air pumps to be controlled independently, the Micro-SD interface provides a means of recording sensor data, and the UART and SWD interfaces improve debugging and firmware-development efficiency. The Electrical Rules Check returned zero errors and zero warnings, and the schematic, footprint assignment, PCB layout, and 3D renders collectively confirm that the design intent has been correctly translated into a physical, manufacturable board layout.

From an engineering standpoint, particular attention was given to power integrity, analog signal quality, high-current actuator routing, grounding, decoupling, and design-rule compliance, with clear separation maintained between sensitive sensor circuitry and pump-switching circuitry to preserve measurement reliability. Overall, the design demonstrates the successful integration of multiple embedded-system functions into a single PCB and provides a strong foundation for a practical environmental-monitoring and automated water-control system. Further hardware bring-up, sensor calibration, and functional testing are required to establish the final quantitative performance of the completed board.